Date:

LAB # 05

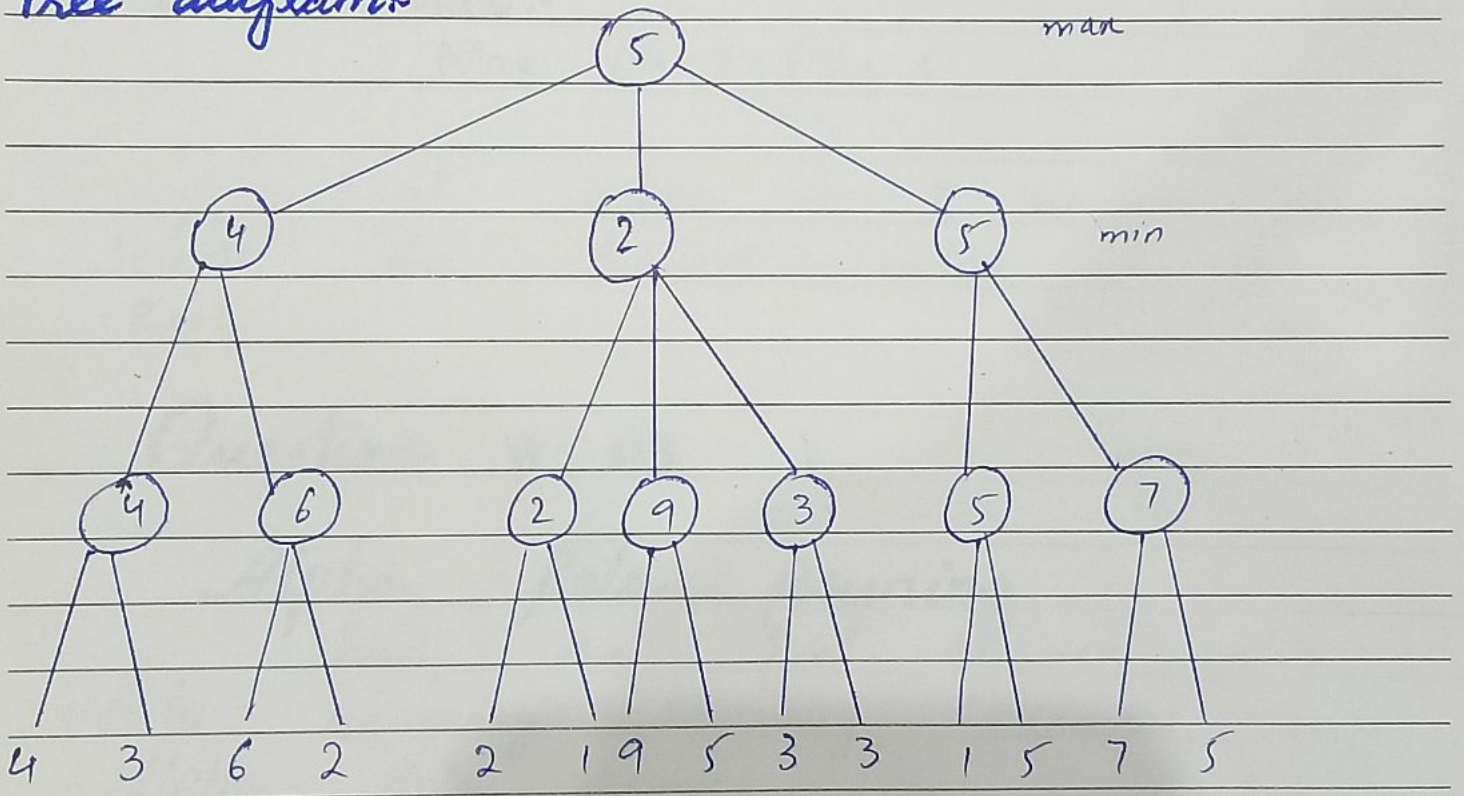
Task 1

Minimax Algorithm

Objectives-

Find the Final value of the root node using the Minimax algorithm

Tree diagram:-



Max nodes (Bottom to top
Max selects highest value

Value	Max
4, 3	4
6, 2	6
2, 1	2
9, 5	9

Value	Max
3, 3	3
1, 5	5
7, 5	7

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MIN Nodes:-

Min nodes selects the lowest values

A = (6,4) ⇒ 4
B = (2,9,3) ⇒ 2
C = (5,7) = 5

Root Max:-

Max (4,2,5) ⇒ 5

Final value is 5 because Max selects the highest value at the Root.

Question # 03

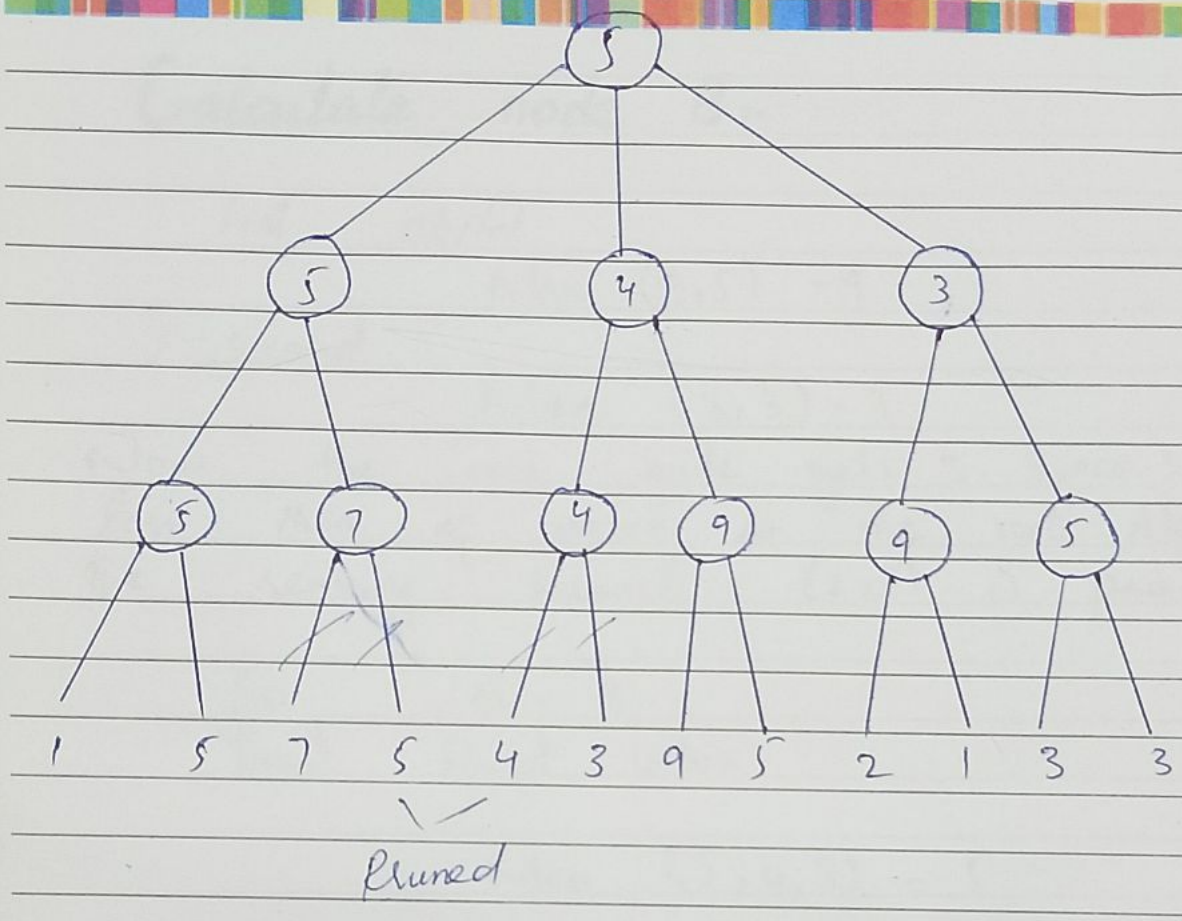
Alpha - Beta pruning

Find the final Minimax value while removing unnessary branches using Alpha - Beta pruning.

For better pruning the visiting order is changed to C, A, B

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Calculate node C:-

$$C = \text{Min}(\text{Max}(1,5)\ \text{Max}(7,5))$$

$$C = \text{Min}(5,7) = 5$$

Calculate node A:-

First child

$$\text{Max}(4,3) = 4$$

Now

$$\alpha = A \quad , \quad B = 4$$

Since $\alpha \geq B$

So $A = 4$

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Calculate node B:-

Fisst child

Max (9, 5) = 9

Second

Max (3, 3) = 3

Now the min node gets 3. Since 3 is less then of equal to the root's Alpha (5) The remaing branch (2, 1) is pruned

So B = 3

Final Root value

Max (5, 4, 3) = 5